Johnson&Johnson
MedTech

Design Optimization Proposal for Enhanced Robotic Arm Responsiveness

1. Problem Statement

Identified Issues:

1. **Actuator Strain**:
   - **Description:** Excessive strain on actuator causing slower response times to input commands.
   - **Impact:** Slow response times to input commands cause reduced precision, reducing the overall reliability of the robot during surgical procedures.
1. **Sensor Misalignment**:
   - **Description:** Poor sensor positioning causing delays and reduced accuracy of feedback control signals.
   - **Impact:** Inaccurate control signals may introduce an unacceptable amount of steady-state error.

2. Proposed Modifications

Modification 1:

- **Proposal**: Optimize load distribution and replace current actuators with higher load bearing actuators.
- **Justification**: Decreasing the load on the actuators while increasing actuator strength will reduce strain on actuators, leading to quicker response times.

Modification 2:

- **Proposal**: Realign and recalibrate sensors.
- **Justification**: Recalibration will ensure that all input data to the sensors is accurately captured, reducing error and increasing response time.

3. Simulation Results

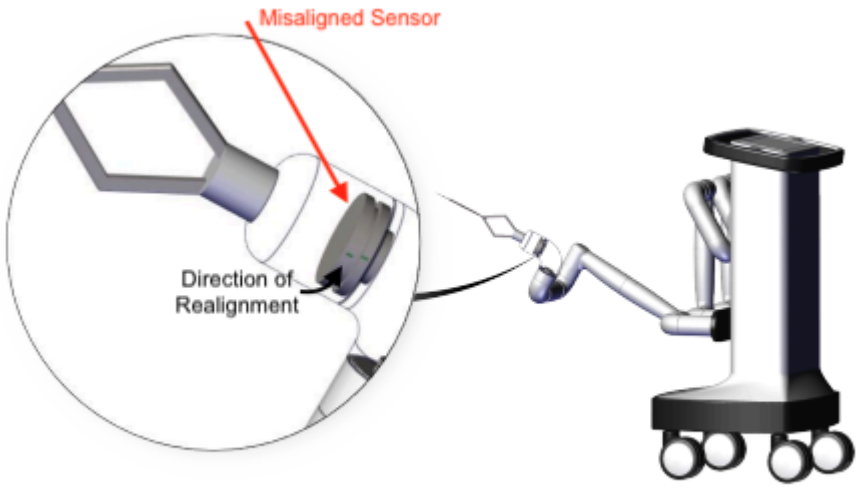
- **Initial Metrics**:
  - **Actuator Durability Score:** 75 (out of 100)
  - **Sensor Feedback Delay:** 0.20 seconds
- **Post-Optimization Metrics**:
  - **Actuator Durability Score:** 99.75 (out of 100)

- **Sensor Feedback Delay:** 0.10 seconds

- **Key Insights**:
  - Utilizing an efficiency improvement factor of 50%, sensor feedback delay was reduced from 0.20 seconds to 0.10 seconds. Utilizing a reinforcement factor of 33% increased actuator durability score from 75 to 99.75.

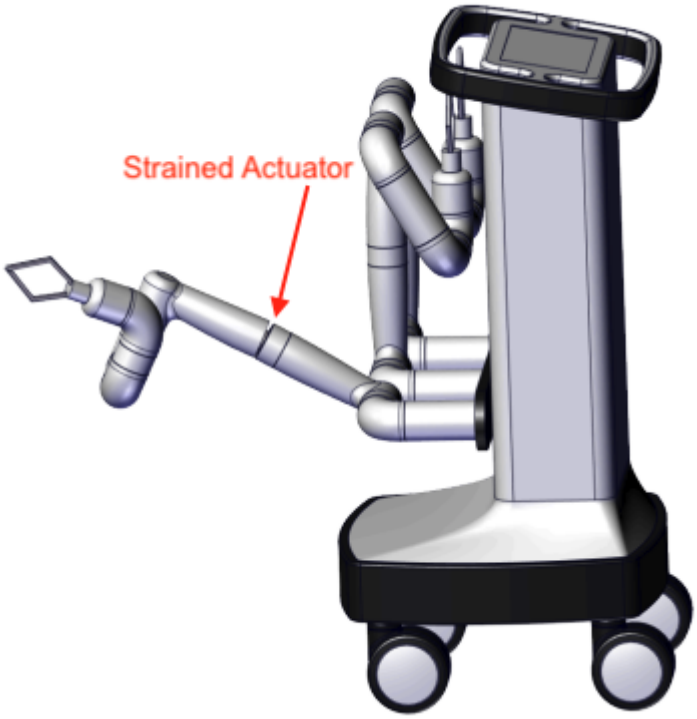
4. Annotated Visual Examples

Visual Example 1: Current Sensor Misalignment vs. Proposed Realignment



- *Brief Description*: Misaligned sensor depicted above, as indicated by the misalignment of the green ticks on both sides of the sensor. Proper alignment is ensured by properly aligning the green tick marks.

Visual Example 2: Actuator Upgrade Impact



- *Brief Description*: High strain on the actuators can cause the joint to bend slightly, reducing accuracy and responsiveness. Optimized load distribution and stronger actuators ensure increased rigidity, leading to improved accuracy and response times.

5. Expected Outcomes

1. **Improved Responsiveness**: 50% increase in sensor efficiency expected to cause a 50% decrease in responsetime, from 0.20 seconds to 0.10 seconds.

2. **Enhanced Reliability**: 50% increase in sensor efficiency and 33% increase in actuator durability expected to cause a significant increase in overall system reliability.

3. **Increased Durability**: 33% increase in actuator durability expected to improve durability score from 75 to 99.75.

6. Conclusion

Summarize the impact of your proposed modifications:

- Replacing actuators with higher load-bearing actuators alongside optimizing load distribution on the actuators resulted in a 33% increase in durability. Realigning/recalibrating sensors resulted in a 50% decrease in response time from 0.20 seconds to 0.10 seconds. By making the system more durable and responsive, system robustness and reliability is increased, ensuring consistent and reliable performance during surgical operations.